Active Inference ~ Parr, Pezzulo, Friston ~ Chapter 10 ~ BookStream #002.03

> **Source**: [YouTube] **Playlist**: Background & Context videos for "Active Inference: The Free Energy Principle in Mind, Brain, and Behavior" **Duration**: 26:10 | **Uploaded**: Unknown **Transcript method**: auto_caption

---

okay this video is going to be an overview of par pulo and friston's 2022 active inference textbook chapter 10 active inference as a unified theory of sentient Behavior I'll read the quote and then Ali please feel free to give your first overview take on the chapter in general we are least aware of what our minds do best Marvin Minsky so what is chapter 10 about okay so chapter 10 is by far my uh most favorite chapter in the whole book but uh uh at the same time uh it differs from all the other chapters um in the sense that it tries to connect connect active inference to so many other disciplines and also uh provides uh some kind of unifying themes in order to tie many of the looser strands uh of the theoretical work that's been done so far at least until um um I mean the books publication so it's a pretty pretty uh enticing chapter in terms of um I mean providing uh many promising uh or introducing some um future research um trajectories or at least promising research uh trajectories for future so uh for anyone who wants to pursue active influence research I believe this particular chapter here can provide uh I mean quite a bit bit of uh interesting ideas to pursue yes this will just be a quick overview they're going to connect a lot of dots they abstract away from the specific active inference generative models discussed in previous chapters and focus on integrative aspects of the framework which as Ali mentions is one of the most exciting and motivating and inspiring Parts we're going to find out by the end of the chapter whether we have what we sought after a unified perspective on problems like perception action selection attention and emotion regulation all these topics which are studied with obviously a diversity of methods but also on the theoretical side essentially in commiserate Frameworks and so active inference here is being positioned as a way to integrate across these different cognitive phenomena and on the more historical or philosophical side how established theories like cybernetics idea motor theory of action reinforcement learning and optimal control can also be understood within the integrative stance that active inference provides section 10.2 wrapping up is like a speedrun on the book itself they Summarize chapter one active inference as a normative approach a chapter with a lot of similarities to chapter 10 chapter two the low road to active inference how basian statistics chapter three the high road to active inference the free energy principle and the imperative to exist chapter four the formal aspects of active inference with a focus on variational inference on generative models chapter 5 they developed from the abstract perspective in chapter 4 and went into more detail on the process theory about how the principle may be implemented by the brain and they gave a bunch of examples of systems of interest and related work on the Maman nervous system chapter six the recipe to design active inference model chapter six kicks off the second half of the book and takes a pragmatic turn towards Hands-On application Learners building their own generative models along the way chapters seven and eight are very curtly summarized here this pair of chapters describes the continuous in chapter eight and discreet in chapter seven generative models and chapter nine on empirical modelbased analysis looks at how to connect data to generative models so any comments on that or continue on with section 103 connecting the dots I think we can go on to chapter to section 10.3 because uh you provided the basic summary of section 10.2 perfectly thanks so what about 10 three okay so 103 um is about going Beyond some uh specific uh aspects of um I mean cognitive facilities of of the agents and try to uh somehow uh integratively model uh the different sub systems of um cognitive facil of the of uh the particular agents um cognition uh so for for instance we can model memory attention perception and uh all all of these uh different aspects of cognition um in this integrated uh formula of active inference within this integrated formalism of active inference and try to uh somehow see the relationships between all of these uh different aspects of cognition and I believe uh this uh integrative work can provide um many Illuminating uh insights about how various aspects of uh given agents cognition is related to each other which is I believe a very important open question in cognitive psychology as well great and the first paragraph references this direction in cognitive science which is rather than to develop a simple or Advanced model of one cognitive phenomena separated out from the others the suggestion of Daniel dennet in 1978 was to model a whole iguana a complete cognitive creature perhaps a simple one but 103 reiterates the way that active inference can be used to connect dots and integrate across cognitive phenomena okay 104 predictive brains predictive minds and predictive processing what would you say about this uh all right so this is also a recurring question uh among the people who uh who were introduced to active inference for the very first time um namely what is uh the distinction between active inference and all the other uh related theoretical Frameworks such as predictive uh predictive processing predictive coding and so on and uh this partic this section uh gives some hint about uh the similarities and also the differences uh between all of these uh different approaches and how they relate to each other whether they're subset um subset of one one of each other or they're distinctly different uh and fundamentally different from each other so um this this section here 10.4 um I believe can be helpful to anyone who wants to understand how active inference fits into this um General framework of recent predicted um predictive way of doing uh cognitive psychology research great and one nice quote here active inference creatures fulfill two imperatives epistemic and pragmatic that's in reference to the expected free energy all right Section 105 perception okay so if we remember from uh chapter 10 uh we were introduced to active inference first by observing how it can model uh the phenomenon uh perception of of an agent um as opposed to uh modeling its action and then we saw that they're basically uh I mean uh they mirror uh image of each other uh the equations that describe those uh phenomenon uh are basically the same equations but just flipped for um the situation we want to model and here uh there's this generalization of how to um how how does it mean to model perception as opposed to uh action and in particular uh what's the basian brain uh hypothesis um I mean what is uh the particular approach taken by basian brain hypothesis uh for modeling uh perception as opposed to some uh other conventional um approaches um to model the same phenomena great 105 perception flows into 10 six action control so these two sections describe the perception and action view of active inference in this action Theory there's subsections on idea motor theory which is a theory with a long history that relates to how cognition influences action cybernetics active inference is closely related to cybernetic ideas and optimal control theory another framework for thinking about action selection these sections differentiate some distinctions of active inference and optimal control theory cybernetics and IDE motor theory however largely as their systems of Interest tend to be overlapping there's more in common with these approaches that students of them will find quite familiar in active inference but again the authors take care to distinguish what active inference does differently than these approaches as well anything else to add there um nothing in particular thanks all right 107 utility in decision- making action expresses priorities Mahatma Gandhi uh okay so here in this section there's also uh this kind of comparative analysis between uh active inference Theory and uh many various a few U different approaches uh used in machine learning so for instance uh we have uh reinforcement learning uh and um I mean various ways of doing um modeling the planning and decision making such as uh the general basian decision Theory uh which is a super set of many different approaches used to model uh the decision making process using basian inference uh and um also there's this planning as inference which is something uh that active inference emphasizes uh uh as again as opposed to some other traditional or conventional uh way of modeling decision making because here as it I mean it's not just simple planning uh as a consequence of uh inference but uh but I mean planning itself is uh considered um I mean the inference uh considered itself as way of uh constructing the U expected free energy uh so uh in this way we can regard planning as inference itself so um this is one of the Novelties of active inference uh compared to any other approaches as also explicitly stated on page all right 108 behavior and bounded rationality okay so uh this section also uh points to some interesting work um with regards to how we can model um different conceptions of rationality and specifically uh bounded rationality and um there are some interesting papers uh which um came out after the publication of uh this book uh in which some some researchers uh pursue this line of research um in order to U model uh even the cons the way people believe conspiracy theories or the way people uh do or make uh apparently irrational decisions and uh they manifest some irrational behavior and so on uh so uh this section points out uh to some tentative ways that active inference can be employed to model uh rationality as well so it's not just about uh modeling perception and action which is a wellestablished way of uh doing AC inference using active inference but from this section onwards uh we uh gradually move towards some Uncharted territories uh which uh I mean admittedly some of them are still uh at their very nent stages so awesome section 109 veilance emotion and motivation okay again uh as I already mentioned uh this is also one of the recent developments of active inference how to model uh emotion with act within active inference framework so I believe uh one of the uh earlier one of the earliest papers that um did this kind of modeling was uh I forgot the title but it's a way of um um of modeling uh an a an agent's behavior in terms of uh the positive or negative veence uh and uh which directly reflects the emotional state of that agent so by agent uh it doesn't necessarily mean a human agent but we can see that this kind of balance in terms of positive and negative balance and how it can uh motivate the behavior of the agent uh can be captured with uh within active inference framework using um using the same techniques we've seen uh for epistemic foraging and so on so uh it's an interesting way of connecting those two subdisciplines yeah I think it's deeply felt affect there's other related words yes that was it again another sort of complex or composite life phenomena moving from emotion onto homeostasis allostasis interceptive processing this section conveys work from variety of authors that's been continued on about homeostasis and allostasis which is the anticipatory steering towards preferred States and section 1011 attention salience and epistemic Dynamics again approaching a sort of cognitive phenomena Nexus and differentiating similar with rule learning causal inference and fast generalization here moving slightly more towards the computational cognitive side anything to say on those sections 1012 1011 or 1010 uh I just wanted to point to uh the way that uh some so up to this point we've seen uh mostly uh the application oriented research of active inference uh in order to deal with u various sorts of scenarios uh I mean from modeling uh the very basic uh cognition of um sentient agents to uh modeling some even social and cultural phenomena uh but I just wanted to point to some uh interesting theoretical work uh that's been going on in parallel with these uh application oriented Works uh which I believe can open up even uh more uh opportunities and uh probably even uh I mean drastically change the view of active inference um in terms of how to how to approach uh different scenarios and different uh phenomena um so uh yeah this kind of philosophical and theoretical uh aspect active inference uh although it's still in its early stages um is beginning to uh lead to some very interesting results uh in terms of applicability of active inference awesome and then section 1013 active inference and other fields open directions as the title would seem to set off this section goes into some of the extensions and elaborations that are quite ongoing in active inference especially related to Social and cultural Dynamics and machine learning and Robotics so subsection one social cultural Dynamics subsection two machine learning and Robotics anything to say on that section uh nothing other than uh there are lots of uh interesting uh um interesting development in this in these uh particular domains uh especially after uh I mean the explosion of llms and generative AI uh we can see the application of uh active inference even to uh these Cutting Edge u u technological developments as well oh and then section 1014 summary over read the Lord of the Rings quote and then look forward to your take on this last section home is behind the world ahead and there are many paths to tread through Shadows to the edge of night until the stars are all a light great so uh obviously this is the last chapter of the whole book and uh personally for me uh reading through this book uh I mean reading reading it multiple times uh have always been uh I mean an increasingly enjoyable Journey for me so each time uh I look at I mean different sections and different uh chapters of the book uh I I feel that there are lots of um interesting potentialities uh that can be uh pursued further to uh and probably perhaps even lead to some uh I don't know some spin-off theories and um some uh Branch to some interesting new Fields uh that is related to active inference but uh they somehow begin to follow the life of their own so uh the uh the main uh strength of this book in my opinion is exactly in providing uh these uh kind of U nent ideas to develop further uh and uh this is in my opinion uh one of the main indicators of a great book uh which uh I mean much more than trying to answer the questions it provides some fertile questions uh in the mind of the reader great points totally concur time and time again we've delighted in this work it's a two paragraph section the first paragraph is worth reading from we started this book by asking whether it is possible to understand brain and behavior from first principles we then introduced active inference as a candidate Theory to meet this challenge we hope that the reader has been convinced that the answer to our original question whether it's possible to understand brain and behavior from first principles is yes we know it side there on how do you feel and then the second paragraph is a quite direct perforation of the fourth wall and they make a really pragmatic day-to-day point about our lives as people learning and applying active inference which is it is important to emphasize that active inference is not something that can be learned purely in theory we encourage anyone who has enjoyed this book to think about pursuing it in practice important writes a passage in theoretical neurobiology and eventually so many other fields too are trying to write down a generative model experiencing the frustration when simulations misbehave and learning from violations of your prior belief when something unexpected happens and this is something that comes up all the time there's so many incredible highflying philosophical questions and in the abstract quite philosophical they are however once there's a generative model on the table once chapter six has been engaged with a lot of times the abstractions and generalizations are left out and you can actually say well how is attention being dealt with in this generative model or what other way could it be done and so making the generative model is a huge moment and they emphasize that that's how you can learn by doing but it's not the only way that you can pursue and engage active inference dayto day it could be in the time interval that's subliminal and between iades it could be having what you expect and prefer for dinner it could be allostasis embedded extended enacted and cultured allostasis and the authors are confident which is to say precise in their beliefs that we will continue to pursue active inference in some form any closing thoughts on 10 ali uh I just wanted to attest um to the last sentence of the book uh because uh definitely after U getting uh familiar uh more and more with different aspects of active inference I can see it um I mean realized or uh or to uh to say it in a better terms I can see it relevant uh to many many different aspects of life not just uh empirical or scientific research but every aspect of life awesome all right well thank you for watching this overview video or playlist and we hope to see you engaged in a textbook group thank you thanks